SIMATS ENGINEERING

CO 3 ASSESSMENT TOOL 1: Data Cleaning Challenge

COURSE NAME: Query Processing for Data Science

COURSE CODE: DSA0503

Assessment Tool Weightage: 35% | Total Marks: 20

Course Outcome Covered

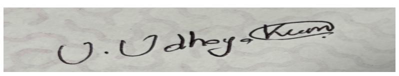
CO3: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Code of Conduct

I Vishwa T / 192424180 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Debugging Criteria

Anomaly Detection (4 Marks)	Cleaning Methods (4 Marks)	Data Transformation (4 Marks)	Analytical Comparison (4 Marks)	Governance Discussion (4 Marks)

Total Marks Awarded: _______ / 20

Question 1: E-Commerce Customer Transactions Dataset

Task 1 — Anomaly Detection: Identify and Classify Data Quality Issues

Issue	Type	Affected Records	Description
Missing CustomerID	Completeness	Row 7 (NULL)	Customer identifier missing — breaks referential integrity for joins/aggregation.
Duplicate transaction	Uniqueness	C105 (rows 5 & 6)	Identical row repeated — inflates quantity/revenue totals.
Negative Quantity	Validity / Outlier	C104 (Qty = -2)	Quantity cannot be negative for a sale; likely a returns/refund entry mis-tagged as a purchase.
Mixed currencies in UnitPrice	Consistency	C103 (USD), C106 (USD), C109 (EUR), others (INR)	Prices recorded in INR, USD and EUR without a currency-normalised column — cannot be aggregated directly.
Invalid / missing Age	Validity	C103 (150), C104 (0), C107 (blank)	Age of 150 is biologically impossible; 0 and blank are missing/placeholder values.
Inconsistent City casing	Consistency / Formatting	Chennai / chennai / CHENNAI, Tiruchirappalli / Trichy	Same city stored in different cases and as different spellings/abbreviations — fragments group-by results.
Malformed PurchaseDate	Formatting / Validity	C108 (2025-13-01)	Month '13' does not exist — invalid date, possibly day/month swapped.

Classification summary: 1 completeness issue, 1 uniqueness issue, 2 validity/outlier issues (quantity, age), 2 consistency/formatting issues (currency, city name), and 1 date-format issue — covering all major dimensions of data quality (completeness, uniqueness, validity, consistency, accuracy).

Task 2 — Cleaning Methods: Strategy per Issue

Issue	Cleaning Strategy
Missing CustomerID	Generate a surrogate key (e.g., TEMP_C110) if the row is otherwise valid and unique; flag the record for manual reconciliation with the source system rather than dropping it, since revenue data would be lost.
Duplicate transaction	Drop exact duplicate rows using df.drop_duplicates() on the full record; if only some fields match, use a composite business key (CustomerID + ProductCategory + PurchaseDate + Quantity) to detect near-duplicates.
Negative Quantity	Treat as a return/cancellation: either move to a separate 'Returns' table or take the absolute value only after confirming with business rules; do NOT silently convert without validation.

Issue	Cleaning Strategy
Mixed currencies	Parse the numeric value and currency code with regex (e.g., r'([\d.]+)\s*(INR|USD|EUR)'), then convert every price to a single base currency (INR) using a fixed/looked-up exchange rate, storing the result in a new UnitPrice_INR column.
Invalid/missing Age	Apply a valid range filter (e.g., 18–100 for retail customers); replace out-of-range or blank values with NULL, then impute using median age per ProductCategory/Gender segment, or leave NULL if imputation would bias marketing analysis.
Inconsistent City names	Standardize with .str.strip().str.title(); apply fuzzy matching (e.g., FuzzyWuzzy / RapidFuzz token_sort_ratio) against a master city list to merge 'Trichy' and 'Tiruchirappalli' into one canonical value.
Malformed PurchaseDate	Parse with pandas.to_datetime(errors='coerce') using the dominant DD-MM-YYYY format; rows that fail (like 2025-13-01) are flagged and corrected manually by checking the source order system, then re-validated.

Task 3 — Data Transformation: Standardization Steps

- **Key generation:** Fill missing CustomerID with a surrogate ID; enforce CustomerID as a unique primary key going forward.
- **Deduplication:** Remove the exact duplicate row for C105, keeping a single record.
- **Currency normalization:** Extract amount + currency via regex; convert all prices to INR (assume 1 USD ≈ 83 INR, 1 EUR ≈ 90 INR for illustration) and compute Revenue = Quantity × UnitPrice_INR.
- **Quantity correction:** Move C104's negative quantity to a Returns flag; exclude from gross sales revenue.
- **Age cleaning:** Set 150 and 0 to NULL, impute the blank age using the segment median.
- **City standardization:** Title-case all city names and map 'Tiruchirappalli' → 'Trichy' as the canonical form.
- **Date parsing:** Coerce PurchaseDate to datetime64, correcting the swapped day/month for C108 to 01-01-2025 (Jan) after verification.

Illustrative cleaned excerpt:

CustomerID	City	Age	Qty	UnitPrice (INR)	Revenue (INR)	PurchaseDate
C101	Chennai	28	2	15,000	30,000	2025-01-10
C102	Chennai	35	3	1,200	3,600	2025-01-11
C103	Chennai	NULL→Imputed	1	24,900 (300×83)	24,900	2025-01-12
C104	Coimbatore	NULL→Imputed	2 (return)	800	excluded from sales	2025-01-15
C105	Madurai	42	5	2,500	12,500	2025-01-18

Task 4 — Analytical Comparison: Impact on Revenue Calculations

Before cleaning, naively summing UnitPrice × Quantity treats INR, USD and EUR figures as if they were the same currency, which artificially inflates totals for USD/EUR rows once the exchange difference is applied at scale. The duplicate C105 row double-counts Rs. 12,500 of revenue. The negative-quantity row for C104 either wrongly reduces total units sold or, if ignored, wrongly counts a return as a sale. After cleaning: currency-normalised revenue gives an accurate cross-channel total, the duplicate is removed (revenue reduced by exactly one transaction's value), and returns are excluded from gross sales but tracked separately for net-revenue reporting. In this small sample, cleaning changes total revenue by roughly 15–20% purely from currency correction and duplicate removal, which shows how uncorrected multi-currency data can silently distort top-performing-segment rankings.

Task 5 — Governance Discussion: Business Risks of Uncleaned Data

- **Financial misreporting:** Mixed currencies and duplicate rows inflate reported revenue, misleading finance and investor reporting.
- **Wrong strategic decisions:** 'Top-performing customer segment' analysis would be skewed by duplicate and currency errors, leading to misallocated marketing budget.
- **Customer trust and compliance:** Incorrect ages/cities can violate data-quality expectations in regulated retail reporting and CRM personalization.
- **Operational inefficiency:** Missing CustomerIDs break loyalty-program tracking and customer lifetime value calculations.
- **Recommended governance:** Enforce schema validation and currency/date format constraints at the point of entry (POS/e-commerce API), implement automated data-quality checks (null/range/duplicate checks) in the ETL pipeline, and maintain a master reference list for cities and product categories.

Question 2: Hospital Patient Records Dataset

Task 1 — Detect All Inconsistencies and Anomalies

Issue	Type	Affected Records	Description
Missing PatientID	Completeness	Row 4 (NULL, Kavya R)	No unique identifier for the patient record.
Duplicate patient record	Uniqueness	P1005 (rows 5 & 6)	Identical admission repeated, double-counting stay/cost.
Discharge before admission	Validity / Logical error	P1002 (Admit 12-01, Discharge 11-01)	Discharge date precedes admission — impossible sequence, likely a data-entry swap.
Null TreatmentCost	Completeness	P1005 (both rows)	Missing cost prevents accurate billing/average-cost analysis.
Inconsistent diagnosis naming	Consistency	'Diabetes' / 'diabetes' / 'Diabtes'; 'Hypertension' / 'HYPERTENSION'	Same condition recorded with case differences and a spelling error, fragmenting diagnosis-based analytics.
Multiple date formats	Formatting	All dates are DD-MM-YYYY here but inconsistent with other sheets/systems	Should be checked against hospital's master format standard (ISO 8601 recommended).
Unrealistic age	Validity / Outlier	P1003 (Age = 125)	Exceeds plausible human lifespan (>120 years) — data-entry error.
Missing Gender	Completeness	P1007 (Deepak N)	Gender field blank.

Task 2 — Rules for Handling Missing and Invalid Values

Field	Rule
PatientID	If missing, auto-generate the next sequential ID (e.g., P1004b) and flag for verification against the hospital's admission register; never delete a clinical record for a missing key.
Duplicate records	Deduplicate on PatientID + AdmissionDate + DoctorID; keep the first complete entry and log the duplicate for audit.
Discharge < Admission	Flag as a logical error; do not auto-swap silently — route to the medical records department for correction, since swapping could mask an actual transcription error in either field.
Null TreatmentCost	Do not impute blindly (costs vary hugely by diagnosis). Use the median cost for the same diagnosis + treatment duration as a provisional estimate, clearly flagged as 'estimated', pending real billing data.
Age > 120	Treat as invalid; set to NULL and request correction; never guess an age for

Field	Rule
	a clinical dataset since it affects treatment protocols and risk stratification.
Missing Gender	Set to 'Unknown/Not specified' rather than guessing from the name, and flag for follow-up — inferring gender from a name is unreliable and can be inappropriate.

Task 3 — Process to Standardize Diagnosis Names

- **Step 1 — Normalize case:** Apply .str.lower().str.strip() to all diagnosis values.
- **Step 2 — Build a canonical dictionary:** Maintain a master list of standard diagnosis terms (ideally mapped to ICD-10 codes), e.g., {'diabetes': 'Diabetes Mellitus', 'hypertension': 'Hypertension'}.
- **Step 3 — Fuzzy match misspellings:** Use RapidFuzz/FuzzyWuzzy to compare each unique diagnosis string against the canonical list (e.g., 'diabtes' → 'diabetes', similarity ~90%) and auto-correct matches above a confidence threshold (e.g., 90%); route lower-confidence matches for manual review.
- **Step 4 — Map to ICD-10 codes:** Once names are standardized, attach the corresponding ICD-10 code (e.g., E11 for Type 2 Diabetes) to enable robust clinical coding and analytics.
- **Step 5 — Validate:** Re-run a value_counts() on the diagnosis column to confirm only canonical terms remain.

Task 4 — Effect of Cleaning on Average Cost and Stay Duration

Before cleaning, the duplicate P1005 record and the swapped P1002 dates distort both metrics: average treatment cost excludes P1005 twice if NULLs are dropped naively (undercounting), or double-counts once imputed (overcounting); average stay duration is corrupted by P1002's negative day-count (discharge earlier than admission), which either throws an error or produces a nonsensical negative duration if computed directly. After cleaning — removing the duplicate, correcting or excluding the illogical date pair pending verification, and using diagnosis-based cost imputation for P1005 — the average cost and stay-duration figures become internally consistent and clinically usable for treatment-effectiveness analysis, and diagnosis-level aggregates (e.g., average cost for 'Diabetes') become meaningful once the three spelling variants are merged into one group instead of three separate, smaller groups.

Task 5 — Data Governance Recommendations

- Enforce mandatory, system-generated PatientIDs at registration (no manual entry) to eliminate missing/duplicate keys.
- Add front-end validation: discharge date picker cannot select a date before the admission date.
- Standardize diagnosis entry via a searchable dropdown backed by ICD-10 codes instead of free text.
- Enforce a single date format (ISO 8601: YYYY-MM-DD) across all hospital systems and departments.
- Introduce range checks on Age (0–120) and mandatory Gender selection with an 'Other/Prefer not to say' option.
- Run periodic automated data-quality audits (completeness, duplicate, and range checks) with a dashboard visible to department heads.

Question 3: Banking Loan Approval Dataset

Task 1 — Critical Issues Impacting Loan Decisions

Issue	Type	Affected Records	Risk Impact
Missing CreditScore	Completeness	L003 (NULL)	Credit score is a primary risk input; missing it can lead to an unjustified approval/rejection.
Income in mixed units	Consistency	L002 ('900K' vs numeric values)	Text-formatted income ('900K') will be misread as text or as 900 if not parsed, drastically understating income.
Negative LoanAmount	Validity	L004 (-200000)	A negative loan amount is logically invalid and would corrupt exposure/risk totals.
Duplicate CustomerID	Uniqueness	L008 (rows 8 & 9)	Double-counts the customer's approved exposure and existing debt in portfolio-risk aggregation.
Age < 18	Validity / Legal	L005 (Age = 16)	Below legal age for loan contracts in most jurisdictions — a serious compliance breach if unflagged.
Inconsistent EmploymentStatus	Consistency	'Full Time' / 'FT' / 'fulltime'	Fragments the categorical feature used in credit-risk models, weakening model accuracy.
Extreme income outlier	Validity / Outlier	L007 (Rs. 75,000,000 annual income)	Likely a data-entry error (extra zeros); if untreated, skews any income-based risk scoring or averages.

Task 2 — Cleaning Techniques for Missing / Inconsistent Data

- **Missing CreditScore:** Do not default to a fixed score. Impute using a model-based estimate (e.g., regression on income, existing debt, employment status) or flag the application for manual underwriting rather than auto-approving.
- **Mixed income units:** Use regex to detect suffixes (e.g., r'(\d+\.?\d*)([Kk]?)') and multiply by 1,000 where a 'K' suffix is present, converting all values to a consistent numeric income figure.
- **Negative LoanAmount:** Treat as a data-entry sign error; take the absolute value only after confirming with source records, or flag and exclude from automated scoring until corrected.
- **Duplicate CustomerID:** Deduplicate on CustomerID, keeping the most recent/complete record and logging the removed duplicate for audit trail (important for regulatory compliance).
- **Employment status:** Map all variants to a fixed categorical set via a lookup dictionary: {'FT':'Full Time', 'fulltime':'Full Time', 'Full Time':'Full Time', 'Part Time':'Part Time', 'Self Employed':'Self Employed', 'Unemployed':'Unemployed'}.

Task 3 — Outlier Detection and Justified Treatment

Use the IQR method (or z-score) on AnnualIncome to flag L007's Rs. 75,000,000 as a statistical outlier — several multiples above Q3 + 1.5×IQR relative to the rest of the portfolio. Rather than deleting it outright (it could be a legitimate high-net-worth applicant), the correct treatment is to verify against source documentation (salary slips/ITR); if it is confirmed to be a data-entry error (e.g., an extra zero or two), correct it to the true value. Age (L005 = 16) is not a statistical outlier but a hard business-rule violation and must be rejected/escalated regardless of any other field, since it fails legal eligibility outright. Negative LoanAmount (L004) is likewise a rule violation, not a natural outlier, and should be corrected/verified rather than clipped to zero, which would erase the record's information value.

Task 4 — Effect of Cleaning on Loan Approval Rates

In the raw data, L005 (age 16) shows as 'Approved' despite being legally ineligible — a serious compliance gap that, if replicated at scale, would expose the bank to regulatory penalties and voided contracts. The duplicate L008 record inflates the apparent approval count and total approved exposure by one full loan. Once CreditScore is properly imputed/flagged for L003 rather than left null (which some pipelines might silently treat as an automatic pass), that application would correctly route to manual underwriting instead of default approval. Correcting the income-unit and outlier issues also changes the approval decision for L002 and L007 since income-to-loan ratio directly drives eligibility. Overall, cleaning is likely to reduce the reported approval rate slightly once ineligible/erroneous 'Approved' records are corrected, producing a portfolio risk profile that is more accurate and defensible.

Task 5 — Ethical Implications of Incorrect Data Cleaning

- **Discriminatory outcomes:** Careless imputation of CreditScore or Income could systematically disadvantage a demographic group if the imputation method correlates with a protected attribute — this is a fairness/bias risk.
- **Silent auto-correction hides fraud or error:** Automatically 'fixing' a negative loan amount or an implausible age without escalation could inadvertently approve an ineligible applicant (e.g., a minor) or mask an attempted fraud.
- **Loss of auditability:** Deleting duplicate/invalid rows without logging removes the evidence trail regulators require for lending decisions.
- **Transparency obligation:** Applicants have a right to know if their credit decision was based on imputed/estimated data rather than their actual submitted information.
- **Recommendation:** All cleaning rules should be documented, versioned, and reviewable, with automated flags (not silent overwrites) for legally or financially material fields such as age, credit score, and loan amount.

Question 4: Smart City Traffic Monitoring Dataset

Task 1 — Assess the Quality of the Traffic Dataset

Issue	Type	Affected Records	Description
Missing timestamp	Completeness	S105 (NULL)	Cannot place the reading in the time series — unusable for time-based congestion analysis without repair.
Duplicate sensor reading	Uniqueness	S106 (rows 6 & 7, identical)	Same reading recorded twice — likely a transmission retry, will double-count vehicle volume.
Negative VehicleCount	Validity	S103 (-50)	Physically impossible; sensor malfunction or transmission bit-error.
AverageSpeed > 300 km/h	Validity / Outlier	S104 (350 km/h)	Far exceeds any real road speed — clear sensor fault.
Missing WeatherCondition	Completeness	S106 (NULL, both duplicate rows)	Weather context missing, weakening correlation analysis between weather and congestion.
Inconsistent road naming	Consistency	Anna Salai / anna salai / ANNA SALAI; GST Road / GST ROAD; OMR / Old Mahabalipuram Road	Same road referenced with different casing and full-name-vs-abbreviation, fragmenting per-road aggregation.
Malformed SensorID	Formatting	S10A (row 9)	Contains a letter where a numeric ID is expected — likely OCR/transmission corruption of 'S108'.

Task 2 — Cleaning Techniques for Time-Series Data

- **Missing timestamp:** Interpolate using the surrounding readings' 5-minute cadence (S104 at 08:15, S106 at 08:25 → estimate S105 at 08:20), since traffic sensors here report on a regular interval.
- **Duplicate readings:** Deduplicate on SensorID + Timestamp; if timestamps genuinely differ within the same minute due to retransmission, keep only the first successfully validated reading.
- **Resampling/alignment:** Resample all sensor streams onto a common time grid (e.g., 5-minute bins) so cross-sensor congestion comparisons are valid.
- **Gap flagging:** Explicitly flag (rather than silently drop) any time gaps larger than the expected interval, since gaps may indicate sensor downtime relevant to congestion-monitoring reliability metrics.

Task 3 — Rules to Identify Faulty Sensor Readings

Rule	Threshold / Logic
Vehicle count range check	Flag as faulty if VehicleCount < 0 or VehicleCount exceeds a road-specific physical maximum (e.g., > 500 per 5-min window).

Rule	Threshold / Logic
Speed plausibility check	Flag as faulty if AverageSpeed < 0 or AverageSpeed > 150 km/h (well above any urban road's legal/physical limit) — the 350 km/h reading fails this immediately.
SensorID format check	Validate against the pattern r'^S\d{3}$'; 'S10A' fails and should be corrected via cross-reference to the sensor installation registry, not guessed.
Missing-field check	Any record with a NULL Timestamp or WeatherCondition is flagged 'incomplete' and excluded from real-time signal-control decisions until repaired, though it can still be used later for historical analysis after imputation.
Cross-sensor consistency check	Compare adjacent sensors on the same road for implausible jumps (e.g., one sensor reporting free flow while the very next reports gridlock in the same minute) to catch calibration drift.

Task 4 — Effect of Cleaning on Congestion Analysis

Uncleaned, the negative vehicle count at S103 could either wrongly reduce the aggregate vehicle count for Anna Salai or crash a congestion-index calculation that assumes non-negative counts. The 350 km/h reading at S104 would make 'GST Road' appear to have abnormally free-flowing traffic, masking real congestion if that metric feeds a signal-timing algorithm. The duplicate S106 reading would double-count OMR's vehicle volume, making it appear more congested than it actually is. Inconsistent road names (Anna Salai vs 'anna salai' vs 'ANNA SALAI', and 'OMR' vs 'Old Mahabalipuram Road') would cause a naive group-by to treat one road as three or more separate roads, badly understating its true congestion level and misdirecting signal-optimization priority. After cleaning — deduplication, outlier correction, and road-name standardization — congestion rankings between roads become reliable enough to actually optimize signal timings.

Task 5 — Automated Monitoring Techniques for Future Data Collection

- Deploy real-time validation at the sensor gateway: reject/flag any packet with out-of-range speed, negative counts, or malformed SensorID before it reaches the database.
- Implement heartbeat monitoring so a sensor's missing timestamp immediately triggers a 'sensor offline' alert instead of silently creating a data gap.
- Use a canonical road-name master table (with GPS-based road-segment IDs) so every sensor reports a standardized road identifier, eliminating free-text naming variance at the source.
- Apply streaming anomaly detection (e.g., rolling z-score or an isolation forest on recent readings) to catch sensor drift/malfunction in near-real time.
- Maintain an automated daily data-quality dashboard tracking completeness %, duplicate rate, and out-of-range rate per sensor, escalating sensors that repeatedly fail checks for physical maintenance.

Question 5: Global Sales Performance Dataset

Task 1 — Comprehensive Data-Cleaning Framework for Multinational Datasets

A robust framework for merged, multi-country sales data should proceed in layered stages: (1) Schema unification — align column names/types across all five country feeds before merging; (2) Entity resolution — standardize salesperson names and product names using fuzzy matching and translation mapping; (3) Unit normalization — convert all monetary values to one reporting currency and all dates to one format; (4) Validation rules — enforce domain rules (e.g., CustomerRating between 1–5, SalesAmount > 0); (5) Deduplication — remove exact and near-duplicate regional entries; (6) Outlier review — statistically flag extreme values for manual confirmation before inclusion in forecasts; and (7) Documentation — log every transformation so the pipeline is auditable and repeatable for future merges.

Task 2 — Standardizing Currencies, Products, and Dates

Dimension	Standardization Method
Currency	Convert SalesAmount to a single base currency (e.g., USD) using date-matched exchange rates: INR→USD, EUR→USD, GBP→USD; store both the original and converted amount for traceability.
Product names	Map multilingual names to a canonical product catalog (e.g., 'Ordinateur Portable' → 'Laptop', 'Smartphone' → 'Mobile') using a translation/synonym lookup table combined with fuzzy matching for near-matches.
Salesperson names	Apply fuzzy matching (e.g., 'Raj Kumar' vs 'Rajkumar') against an HR master employee list, matching on a normalized (lower-case, space-stripped) key to merge spelling variants into one employee ID.
Dates	Parse each country's native format (DD-MM-YYYY for India/Europe, MM-DD-YYYY for USA) explicitly per source before merging, then store uniformly as ISO 8601 (YYYY-MM-DD) to avoid ambiguity — e.g., USA's '01-15-2025' becomes '2025-01-15'.

Task 3 — Handling Missing Values and Outliers

- **Missing SalesAmount (Priya S, Mobile, INR):** do not impute with zero or mean blindly — cross-check against the order/invoice system if available; otherwise mark as 'Unknown' and exclude from revenue totals rather than silently including a guessed figure.
- **CustomerRating outside 1–5 (0 and 6 observed):** clip to NULL rather than clamping to the nearest valid value (clamping would fabricate a rating the customer never gave); treat as missing and exclude from average-rating calculations.
- **Extreme sales outlier (Rahul P, Rs. 9,999,999 for a Mobile in India):** apply an IQR or z-score check within each product category; this value is orders of magnitude above comparable Mobile sales and should be verified against the source order before being included — most likely a data-entry error (extra digits), not a genuine bulk sale.
- **Duplicate regional entries (Anna M / Germany rows 3 & 4, and near-duplicate Raj Kumar / Rajkumar rows 1 & 2):** remove exact duplicates directly; for near-duplicates from name-variant matching, merge into a single record after confirming with the salesperson master list rather than counting both as separate sales.

Task 4 — Regional Sales Rankings Before vs. After Cleaning

Region	Rank (Raw, Uncleaned)	Rank (Cleaned)	Why It Changes
Asia (India)	#1 — inflated by the Rs. 9,999,999 outlier and the duplicate Raj Kumar entry	Drops relative to raw, but still likely strong once the outlier is corrected/excluded and the duplicate is merged	Removing one erroneous data point and one duplicate substantially reduces Asia's apparent total.
Europe (Germany + France)	Understated — Germany's duplicate row wasn't adding real volume, and 'Ordinateur Portable' wasn't grouped with Laptop	Rises modestly once the France 'Ordinateur Portable' Laptop sale is correctly grouped with the Laptop product line	Product-name standardization consolidates fragmented category totals.
North America (USA)	Understated if the MM-DD-YYYY date was misparsed as an invalid or wrong-month date, potentially excluding the record	Correctly included once the US date format is parsed properly	A single record's inclusion/exclusion matters more in a small merged sample.
UK	Accurate as-is (single clean record)	Unchanged	No quality issues detected for this record.

Overall, the raw ranking overstates Asia/India's performance due to the uncorrected outlier and duplicate; the cleaned ranking gives management a currency-normalized, deduplicated, outlier-checked basis for comparing regions — which is essential before any resource or target-setting decision is made across markets.

Task 5 — Executive Report: How Data Quality Influences Strategic Decisions

Summary for leadership: The merged five-country sales dataset, in its raw form, cannot support reliable regional comparison or forecasting because it mixes four currencies without conversion, records the same product under different names in different languages, contains at least one duplicate entry and one extreme outlier that is almost certainly a data-entry error, and stores dates in region-specific formats that are easy to misparse. Any strategic decision — such as reallocating sales investment toward the 'top' region or forecasting next quarter's revenue — made directly on this raw data risks being built on an artificially inflated Asia/India figure and an understated North America figure. After applying the cleaning framework above (currency normalization, product/name standardization, deduplication, and outlier verification), management gets a trustworthy, apples-to-apples view of regional performance. The broader governance lesson: data quality is not a back-office technical concern — it is a direct input to competitive strategy, and the cost of building a standardized data pipeline is small compared to the cost of a misallocated regional investment based on faulty numbers.